

ABINASH SAMBANDHAM

AI Engineering Lead | Generative AI | Agentic Systems | Enterprise AI

Coimbatore, India · Open to Relocation

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SUMMARY

AI engineering leader with 5+ years building enterprise software, including production LLM systems, RAG and agentic workflows for customers across Europe and the US. Hands-on from customer requirements and AI application architecture through LLM orchestration, retrieval, evaluation, observability and production support. Created the **AI engineering framework now standard across the company**, and owned AI development for a retail compliance platform that reached **800+ brands and 15 retailers in 12 months**.

CORE AI ENGINEERING

LLM Applications & Agents

AI Application Architecture | LLM Orchestration | Agentic AI | Multi-Agent Systems | Tool & Function Calling | Structured Outputs | LangGraph | LangChain | Azure AI Foundry Agents | Custom MCP Servers | Prompt & Context Engineering

LLM Providers & Platforms

OpenAI API | Anthropic Claude API | Azure OpenAI | Azure AI Foundry | AWS Bedrock

RAG & Retrieval

RAG | Embeddings | Chunking | Hybrid Search | Re-ranking | Azure AI Search | Chroma | Retrieval Evaluation

Evaluation & Observability

LLM Evaluation Frameworks | Azure AI Foundry Evaluations | Langfuse | AI Tracing | Regression Evaluation Sets | Drift Detection | Human-in-the-Loop Review

Production LLM Systems

Inference Optimisation | Semantic Caching | Model Routing | Guardrails | Schema Validation | Retries & Fallbacks

Document AI & Cloud

Azure Document Intelligence | AWS Textract | Azure API Management | AWS Lambda | AWS S3

Backend Engineering

Python | FastAPI | Java | Spring Boot | Microservices | RabbitMQ | Docker | Kubernetes | Jenkins

Frontend & Data

TypeScript | Next.js | React | Angular | tRPC | Prisma | MongoDB | PostgreSQL | Elasticsearch

PROFESSIONAL EXPERIENCE

TECHNICAL LEAD | AI ENGINEERING & AGENTIC SYSTEMS

TrusTrace · Coimbatore

Apr 2025 – Present

Enterprise supply-chain traceability and compliance SaaS serving customers across Europe and the US.

Team Leadership

- Line-manage **a team of five** on Supplier Data Management (four engineers, one QA) while staying hands-on in design, architecture and development, owning their technical direction, sprint delivery, code review and release quality.
- Provide technical leadership without line management across One Retail Hub and the AI engineering platform, setting architecture, reviewing design and code, and mentoring engineers on AI engineering practice.

One Retail Hub: AI-Driven Retail Compliance Platform

Next.js | TypeScript | Python | Azure AI Foundry | Azure OpenAI | Claude | Kubernetes | Stripe

- Led the end-to-end AI build of One Retail Hub, from requirements and architecture to AI solution design and production delivery, as **one of two lead-level engineers** on the core AI build.
- Built a **multi-stage RAG assessment pipeline** over unstructured compliance documents using Azure Document Intelligence, hybrid search, re-ranking and Claude, with risk-weighted evaluation for EUDR, CSDDD and CSRD. Cut supplier compliance turnaround **from 4 days to 30 minutes** while keeping human-in-the-loop review for auditability.
- Optimised inference cost and latency with **semantic caching, model routing** that sends triage and classification steps to lighter models, and prompt and chunk tuning.
- Implemented **multi-agent data-collection workflows with LangGraph and MCP**, where planning and tool-calling agents handle web and document ingestion. Added **production guardrails**: structured outputs with Pydantic and JSON schema validation on every tool and function call, retries with model and path fallbacks when calls fail, and escalation of low-confidence or failed runs to human review.
- Designed the platform to be **configuration-driven**, enabling customers to extend workflows without customer-specific code.

- Scaled the platform to **800+ brands and 15 retailers within 12 months**, supported by largely self-service onboarding.

Enterprise AI & Customer-Facing Engineering

- Direct enterprise AI delivery from customer requirements and architecture through production, working directly with technical and business stakeholders in the US and Europe.
- Run requirements sessions, technical discovery and AI proofs of concept with enterprise brands and retailers, turning business problems into deployable AI solutions.
- Built and shipped **four customer-facing micro-applications** independently, each taking **about two weeks** from requirements to production.
- Own customer escalations and production support, turning recurring requests into reusable, configurable platform features.

AI Engineering Platform & Agentic SDLC

Python | Claude Code | Cursor | Azure AI Foundry | Langfuse | MCP | Playwright | Jenkins | Chroma

- Architected and built the company's **AI-driven SDLC framework**, connecting specialist agents for requirements refinement, development, testing and code review. It went through several iterations, including spec-driven development, and is now the **standard engineering approach across teams**.
- Developed the context layer that grounds agents in current repository, ticket and architecture state, backed by a company-wide knowledge base covering product, engineering and domain context.
- Set up **three-layer verification for AI-generated code**: Playwright and unit tests, regression evaluation sets that catch behaviour drift when prompts or models change, and agent-assisted review with engineer sign-off before merge.
- Replaced manual QA cycles with automated Playwright regression, moving releases **from quarterly to monthly**.

Supplier Data Management

Java | Spring Boot | Python | FastAPI | Angular | MongoDB | RabbitMQ | Azure OpenAI

- Own SDM architecture and delivery end to end, setting technical direction, code quality standards and release readiness.
- Created configurable UI generation, rule-driven workflows, hierarchical data models and a Python validation framework.
- Enabled **10+ supply-chain data-collection use cases** through configuration alone, with no per-customer implementation code.

SENIOR SOFTWARE ENGINEER

TrusTrace · Coimbatore

Apr 2024 – Apr 2025

Java | Spring Boot | Python | MongoDB | Elasticsearch | RabbitMQ | Kubernetes

- Delivered Supply Chain Mapping and Real-Time Traceability for multi-tier supply-chain networks, including compliance and evidence management.
- Automated certificate and compliance-document capture, classification and validation using AI document extraction.
- Sped up traceability queries through MongoDB data modelling, Elasticsearch search, caching and batch processing.
- Implemented event-driven services on RabbitMQ for asynchronous communication and fault tolerance.

SOFTWARE ENGINEER

TrusTrace · Coimbatore

Feb 2021 – Apr 2024

Java | Spring Boot | MongoDB | Elasticsearch | Angular | AWS Textract | Azure Document Intelligence

- Delivered and evolved a production document-extraction service, moving **from AWS Textract to Azure Document Intelligence** after evaluating rule-based and model-based approaches.
- Developed Spring Boot REST APIs with asynchronous processing, rate limiting and third-party integrations.
- Modelled MongoDB schemas and set up Elasticsearch indexes and aggregation pipelines for search and data access at scale.
- Created a reusable Angular component library with NX and Storybook, and a micro-frontend architecture for independent deployments.

EDUCATION

B.E., Electronics & Communication Engineering

2017 – 2021

Sri Shakthi Institute of Engineering & Technology, Anna University